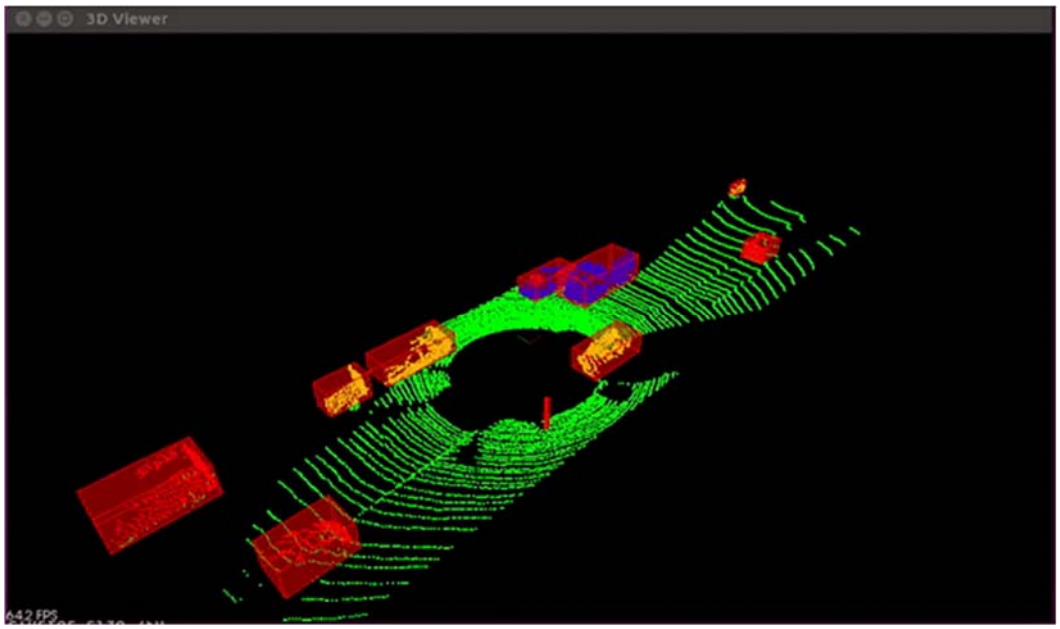
Computer Vision
Lecture 10: Automotive LiDAR Sensor

Yeungnam University
MSP lab.
Ho-Youl JUNG

영남대학교 Yeungnam University

LiDAR

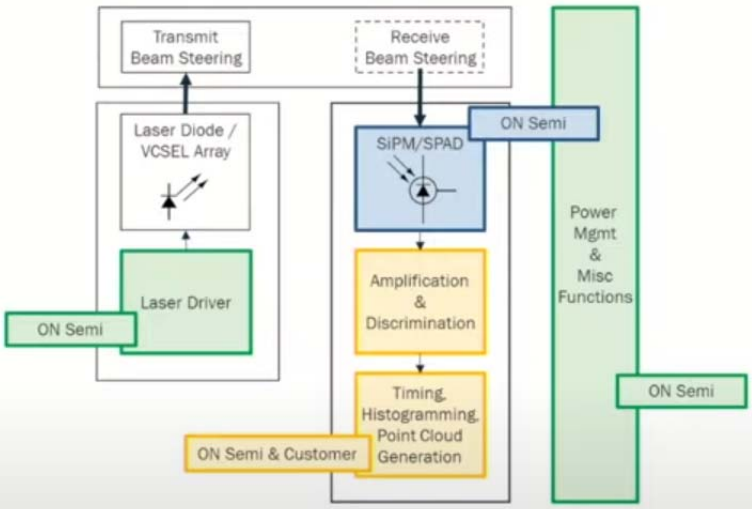
https://www.youtube.com/watch?v=KKf7O4IEofg



LiDAR H/W

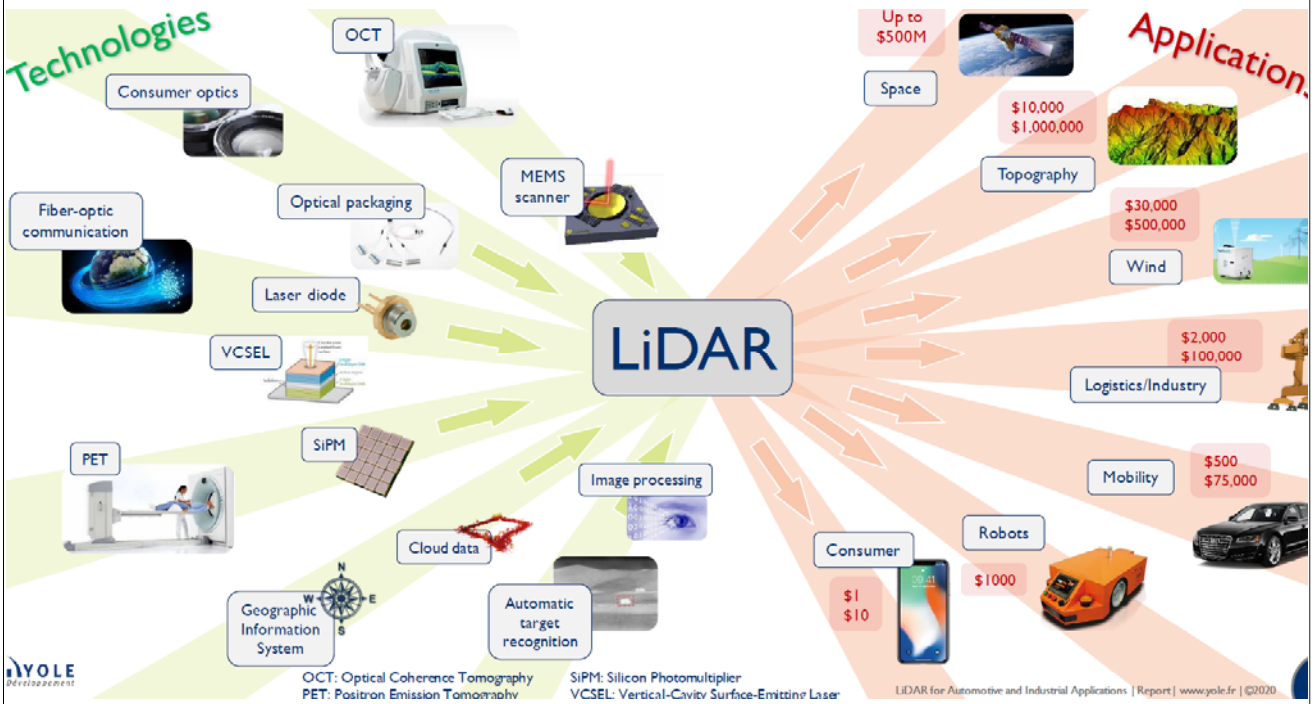
Six Major Hardware Functions block on a LiDAR System:

Transmit	• Illumination in NIR spectrum
Receive	• Return photon detection
Beam Steering	• Scanning/Scene Illumination
Optics	• TX and RX focus
Readout	• Point cloud generation
Power and System	• Power management



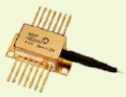
Typical LiDAR System Block Diagram

LiDAR Applications



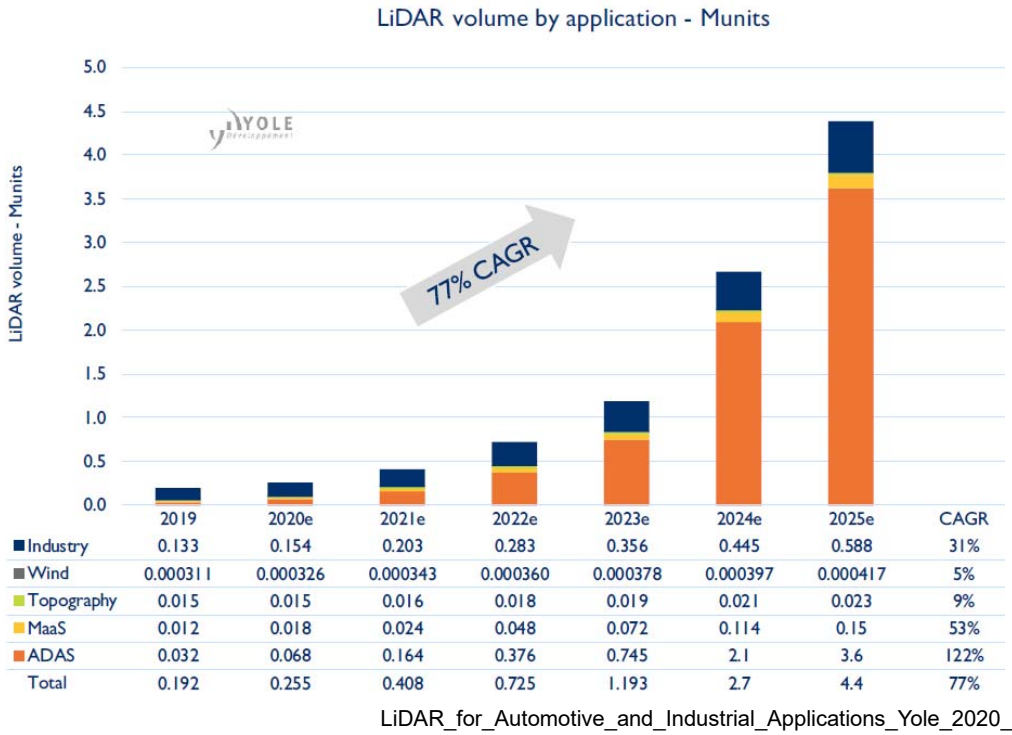
LiDAR_for_Automotive_and_Industrial_Applications_Yole_2020_Report

LiDAR Supply Chain

Supply chain	Sensor Semiconductor	Electronic Module	Sensor System	Platform
LiDAR	 Laser Diodes Photodetectors	 Fiber lasers	 LiDAR	
Computing	 GPU – SoC – SiP	 Computing boards	 AD Computing	

LiDAR_for_Automotive_and_Industrial_Applications_Yole_2020_Report

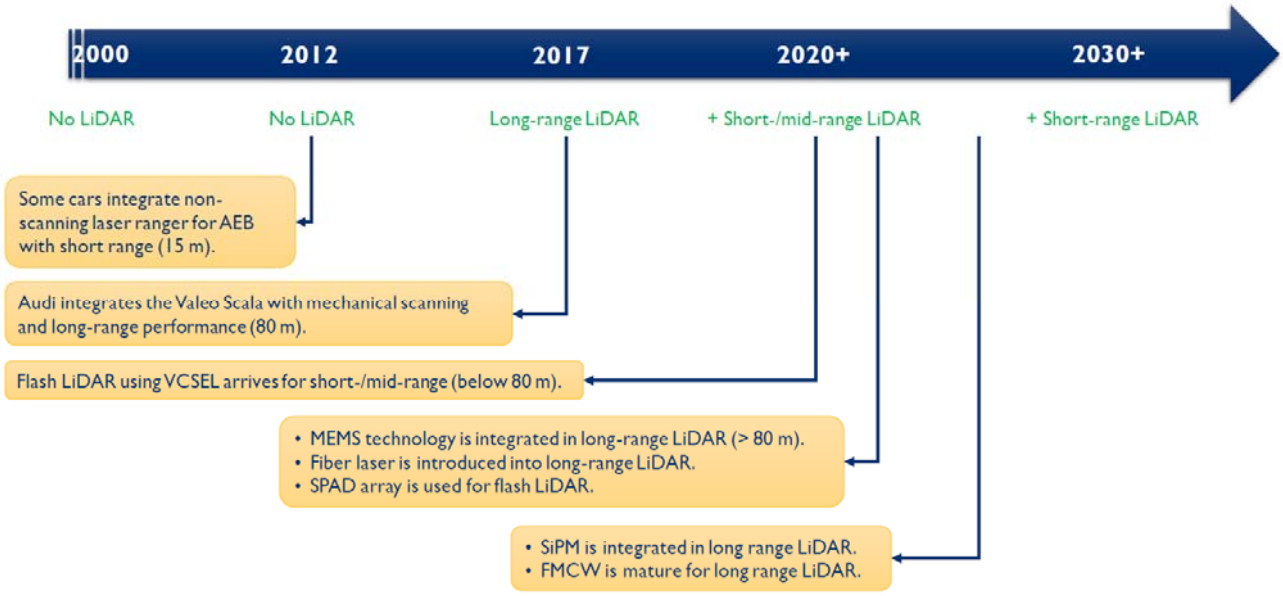
LiDAR Volume Forecast



	2019	2020e	2021e	2022e	2023e	2024e	2025e	CAGR
Industry	0.133	0.154	0.203	0.283	0.356	0.445	0.588	31%
Wind	0.000311	0.000326	0.000343	0.000360	0.000378	0.000397	0.000417	5%
Topography	0.015	0.015	0.016	0.018	0.019	0.021	0.023	9%
MaaS	0.012	0.018	0.024	0.048	0.072	0.114	0.15	53%
ADAS	0.032	0.068	0.164	0.376	0.745	2.1	3.6	122%
Total	0.192	0.255	0.408	0.725	1.193	2.7	4.4	77%

LiDAR_for_Automotive_and_Industrial_Applications_Yole_2020_Report

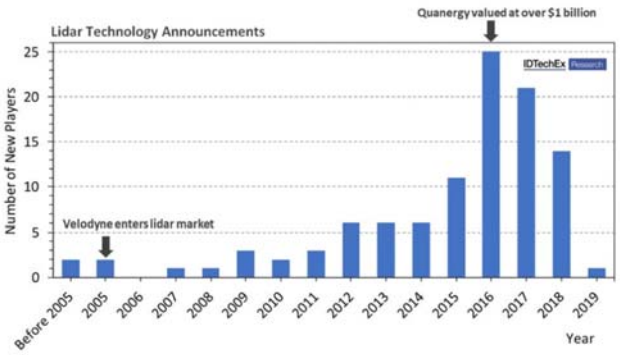
LiDAR Technology Road map

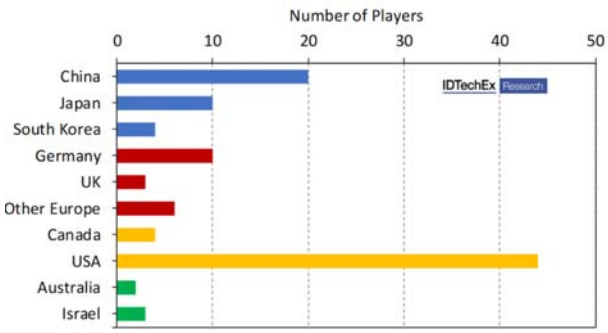


LiDAR_for_Automotive_and_Industrial_Applications_Yole_2020_Report

Automotive LiDAR Players

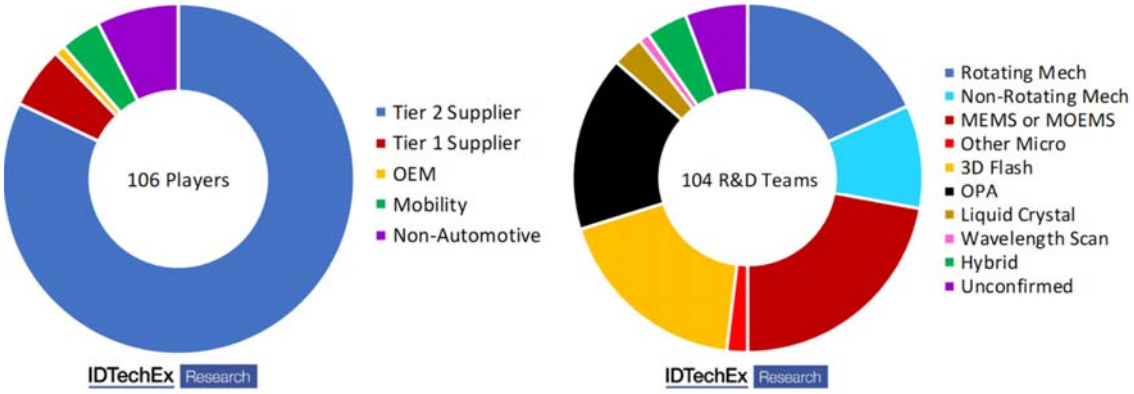
- 107 companies by November 2019,: most companies position their business as automotive Tier 2 suppliers
- 3D lidar players **:** AEye, Baraja, **Blickfeld, Cepton**, **Continental**, **Innoviz, LeddarTech**, Luminar, **Ouster**, **Quanergy**, **SiLC**, TriLumina, **Valeo**, **Velodyne,** and **XenomatiX**, **Opsys**



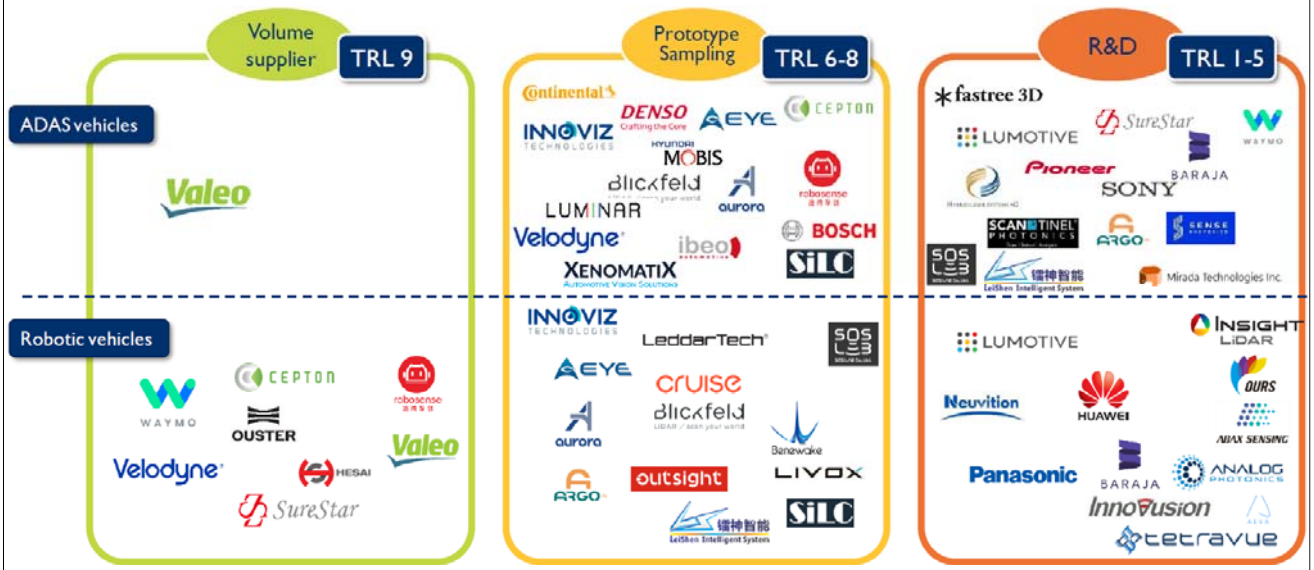


New companies announcing the development of lidar technology for the automotive market. Few companies enter this market today as it is saturated with more than a hundred 3D lidar technologies. Source: "Lidar 2020-2030" IDTechEx report

Automotive LiDAR Players

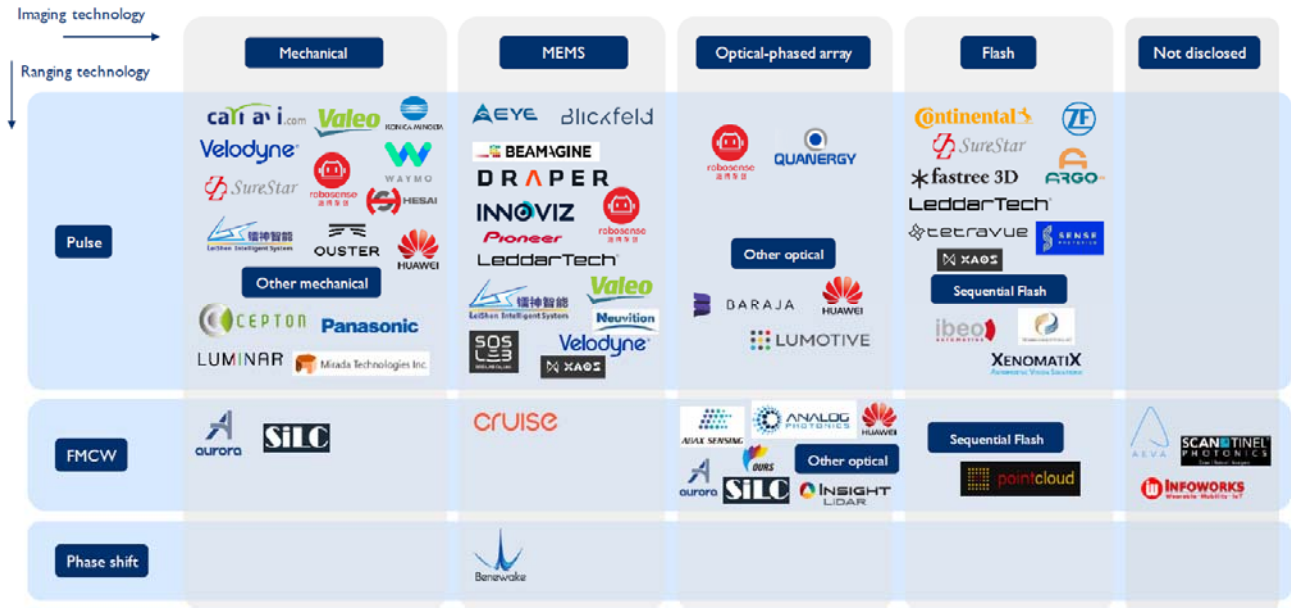


Automotive LiDAR Players



LiDAR_for_Automotive_and_Industrial_Applications_Yole_2020_Report

Technology/Player for Automotive LiDAR



LiDAR_for_Automotive_and_Industrial_Applications_Yole_2020_Report

Automotive LiDAR Players

- 107 companies by November 2019,: most companies position their business as automotive Tier 2 suppliers

- Velodyne (United States)
- Ouster (United States)
- Aeva (United States)
- Luminar (United States)
- Baraja (Australia)
- Innoviz (Isreal)
- Valeo (France)
- Sick (Germany)
- Infineon (Germany)
- Benewake (China)
- Livox (China)
- Denso (Japan)
- Hokuyo (Japan)
- Pioneer (Japan)

- Xenomatix (Belgium)
- Ibeo (Germany)
- Blickfeld (Germany)
- Continental (Germany)
- LeddarTech (Canada)
- Quanergy (United States)
- AEye (United States)
- Robosense (China)
- Hesai (China)

Automotive LiDAR Comparison 1

										
	Valeo ibeo	Velodyne LiDAR	Velodyne LiDAR	Velodyne LiDAR	INNOVIZ	INNOVIZ	LUMINAR	LeddarTech	AEYE	Continental
Model	ScaLa I	Alpha Prime	Velarray	Velabit	InnovizPro	InnovizOne	Hydra	Leddar Pixell	AE110	HFL110v
Application	ADAS/MaaS	MaaS	ADAS	ADAS	ADAS	ADAS	R&D	ADAS	ADAS	ADAS
Channels #	1	128	-	-	MEMS	MEMS	NA	Flash	MEMS	Flash
Range (m) (reflectivity)	80 (6.3%)	220 (10%)	NA	100	80 (10%)	250	250 (10%)	20 (10%)	205 (10%)	Short-range
Points frequency (pts/sec)	14,500	4,800,000	NA	NA	100.000	7,800,000	NA	14,000	18,000,000	120,000
Horizontal FOV (°)	145	360	NA	60	73	125	120	177.5	60	120
Vertical FOV (°)	3.2	40	NA	10	20	25	30	16	30	30
Power (W)	7	22	NA	NA	35	NA	55	20	NA	15
Price ($)	632	NA	NA	100	NA	NA	NA	NA	NA	NA
Status	On sale	On sale	2020	2020	On sale	NA	On sale	On sale	On sale	2020

Automotive LiDAR Comparison 2

										
	robosense	robosense	SureStar	CEPTON	HESAI	HESAI	XENOMATIX	LIVOX	ibeo automotive	LUMOTIVE
Model	RS-LiDAR-M1	RS-Bpearl	C-fans	Vista P90	Pandar64	PandarGT-L60	XenoLidar	Horizon	IBEO NEXT	NA
Application	ADAS	ADAS	ADAS/MaaS	ADAS/MaaS	MaaS	ADAS	ADAS	ADAS	ADAS	ADAS/MaaS
Channels #	MEMS	NA	64/128	NA	64	MEMS	Flash	Risley prisms	Flash	250
Range (m) (reflectivity)	150 (10%)	30 (10%)	200 (10%)	NA	200 (10%)	300 (10%)	200	90 (10%)	150 (10%)	250 (10%)
Points frequency (pts/sec)	1,125,000	576,000	1,280,000	NA	1,152,000	1,700,000	260,000	240,000	NA	9,000,000
Horizontal FOV (°)	120	360	150	NA	360	60	60	81.7	120	120
Vertical FOV (°)	25	90	30	NA	40	20	20	25.1	11	25
Power (W)	25	13	15	NA	22	30	NA	NA		NA
Price ($)	1,898	NA	NA	NA	NA	NA	NA	800	NA	-
Status	On sale	On sale	On sale	On sale	On sale	On sale	Sample	On sale	On sale	POC

Automotive LiDAR Comparison 3

Industry / Topography								
	SICK	SICK	HOKUYO	Benewake	Benewake	blickfeld	blickfeld	OUSTER
Model	LMS500 PRO	MRS6000	UXM-30LN	CE30-A	Tfmini Indoor	Cube Range 1	Cube 1	OS1-128
Channels #	1	24	1	Flash	1 (single point)	MEMS	MEMS	128
Range (m)	26 (10% reflectivity)	30 (10% reflectivity)	30 (10% reflectivity)	4	12	250	250 (10%)	120 (80%)
Points frequency (pts/sec)	114,000	220,000	15,200	150,000	1,000	20,000	130,000	2,600,000
Horizontal FOV (°)	190	120	190	132	3.6	12	70	360
Vertical FOV (°)	-	15	-	9	3.6	10	30	45
Power (W)	22	20	6	6	0.5	15	15	20
Price ($)	12,000	6,500	4,700	800	34.9	NA	NA	18,000
Status	On sale	On sale	On sale	On sale	On sale	Sample	Sample	On sale

Velodyne Velodyne (United States)

- The best known player (American company)
- Basic tech.: 905 nm Wavelength, ToF, Spinning
- In 2005, **HDL-64** rotating mechanical lidar
- In 2017, **VelarrayTM** (micro-lidar array (MLA) technology) for long-range object detection
- In 2019, **VelaDomeTM** for close-range operation
- Recently, **Velabit**, a $100 small lidar offering 60x10 deg at 10-20Hz and 100m@80% reflectively
- **New non-rotational technology** for automotive employs 128 channel VCSEL light source(905nm ToF) and APD detectors, unknown scanning mechanism (probably an electromechanical mechanism)

Velodyne Velodyne (United States)

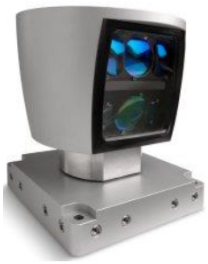
Sensor	HDL-64E	HDL-32	Puck	Puck LITE	Puck Hi-Res	Ultra Puck	Alpha Prime
							
Range	Up to 120m	Up to 100m	100m	100m	100m	200m	Up to 245m[5]
Range Accuracy	Up to ±2 cm (Typical)[4]	Up to ±2 cm (Typical)[1]	Up to ±3 cm (Typical)[1]	Up to ±3 cm (Typical)[1]	Up to ±3 cm (Typical)[1]	Up to ±3 cm (Typical)[1]	Up to ±3 cm (Typical)[1]
# of Lines	64	32	16	16	16	32	128
Horizontal FoV	360°	360°	360°	360°	360°	360°	360°
Vertical FoV	26.9°	41.33°	30°	30°	20°	40°	40°
Horizontal Resolution	0.08° - 0.35°	0.08° - 0.33°	0.1° - 0.4°	0.1° - 0.4°	0.1° - 0.4°	0.1° - 0.4°	0.1° - 0.4°
Vertical Resolution	0.4°	1.33°	2.0°	2.0°	1.33°	0.33° (min)	0.11° (min)
Points Per Second (Single Return Mode)	~ 1,300,000	~ 695,000	~ 300,000	~ 300,000	~ 300,000	~ 600,000	~ 2,400,000
Points Per Second (Dual Return mode)	~ 2,200,000[5]	~ 1,390,000	~ 600,000	~ 600,000	~ 600,000	~ 1,200,000	~ 4,800,000
Refresh Rate	5-20 Hz	5-20 Hz	5-20 Hz	5-20 Hz	5-20 Hz	5-20 Hz	5-20 Hz
Operating Voltage	12V - 32V	9 V - 18 V	9 V - 18 V	9 V - 18 V	9 V - 18 V	10.5 V - 18 V	9 V - 28 V
Power Consumption	60 W (Typical)[2]	12 W (Typical)[2]	8 W (Typical)[2]	8 W (Typical)[2]	8 W (Typical)[2]	10 W (Typical)[2]	22 W (Typical)[2]
Weight (without cabling)	~ 28 lbs. (12.7 Kg)	~1.0 kg	~830 g	~590 g	~830 g	~925 g	~3.5 kg
Operating Temp	-10°C to +60°C[3]	-10°C to +60°C[3]	-10°C to +60°C[3]	-10°C to +60°C[3]	-10°C to +60°C[3]	-20°C to +60°C[3]	-20°C to +60°C[3]
Storage Temp	-40°C to +85°C	-40°C to +105°C	-40°C to +105°C	-40°C to +105°C	-40°C to +105°C	-40°C to +85°C	-40°C to +105°C
Output	UDP packets over Ethernet	UDP packets over Ethernet	UDP packets over Ethernet	UDP packets over Ethernet	UDP packets over Ethernet	UDP packets over Ethernet	UDP packets over Ethernet
Ethernet Connection	100 Mbps	100 Mbps	100 Mbps	100 Mbps	100 Mbps	100 Mbps	1000 Mbps
GPS Timesync	$GPRMC	$GPRMC + $GPGGA	$GPRMC + $GPGGA	$GPRMC + $GPGGA	$GPRMC + $GPGGA	$GPRMC + $GPGGA	$GPRMC + $GPGGA
Laser	903nm Class 1 eye safe	903nm Class 1 eye safe	903nm Class 1 eye safe	903nm Class 1 eye safe	903nm Class 1 eye safe	903nm Class 1 eye safe	903nm Class 1 eye safe
Water Resistance	IP67	IP67	IP67	IP67	IP67	IP67	IP67

63-9645 Rev G

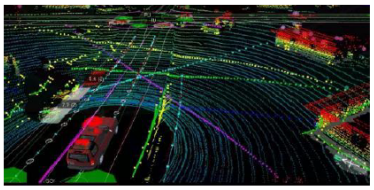
1.Typical accuracy refers to ambient wall test performance across most channels and may vary based on factors including but not limited to range, temperature and target reflectivity. 2. Operating power may be affected by factors including but not limited to range, reflectivity and environmental conditions. 3. Operating temperature may be affected by factors including but not limited to air flow and sun load. 4. Greater than or equal to 80% of channels at ambient wall test; remaining channels better than or equal to 5 cm. 5. Configuration dependent. 6. These are projected specifications for final production parts. The specifications for any sample, prototype, or other non-final or pre-production products may be different from the specifications in this document. For more information, please contact Velodyne Sales.

datasheet

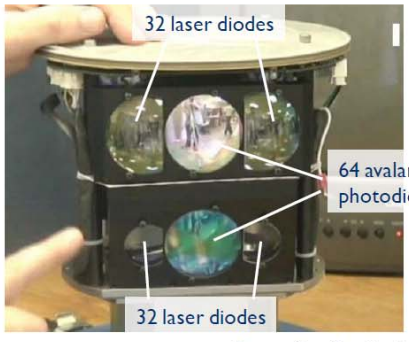
Velodyne HDL-64E



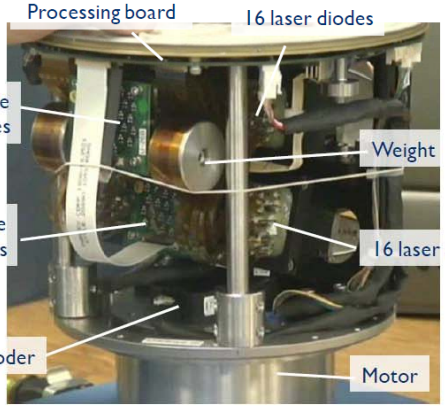
- This is a high spec LiDAR available for automotive.
- The unit price is around $75k (estimated).
- There are 64 channels (vertically).
- Scanning is 360° (horizontally).
- 1.3 million points are measured per second.
- The whole 'head' spins between 5 and 15 Hz.



Front view



Rear view



Velodyne Velodyne (United States)

	Close Range	Directional Sensors		Software
Sensor	VelaDome™	Velarray™	Velabit™	Vella™
				
Range				
Range Accuracy				
# of Lines				
Horizontal FoV				
Vertical FoV				
Horizontal Resolution				
Vertical Resolution				
Points Per Second (Single Return Mode)				
Points Per Second (Dual Return mode)				
Refresh Rate				
Operating Voltage				
Power Consumption				
Weight (without cabling)				
Operating Temp				
Storage Temp				
Output				
Ethernet Connection				
GPS Timesync				
Laser				
Water Resistance				

Visit us for more information
VelodyneLidar.com

Velodyne Velodyne (United States)

- **VelarrayTM** (micro-lidar array (MLA) technology) for long-range object detection
- **VelaDomeTM** for close-range operation
- **Velabit**, a $100 small lidar offering 60x10 deg at 10-20Hz and 100m@80% reflectively

Velarray™

Smart powerful lidar solution for ADAS and Autonomy

With a compact form for seamless integration within a vehicle's body or behind the windshield, the Velarray produces a robust directional image, day or night. Its best-in-class range and resolution allow for faster object identification and longer braking distance at highway speeds. Mass production is slated for 2020.



Velabit™

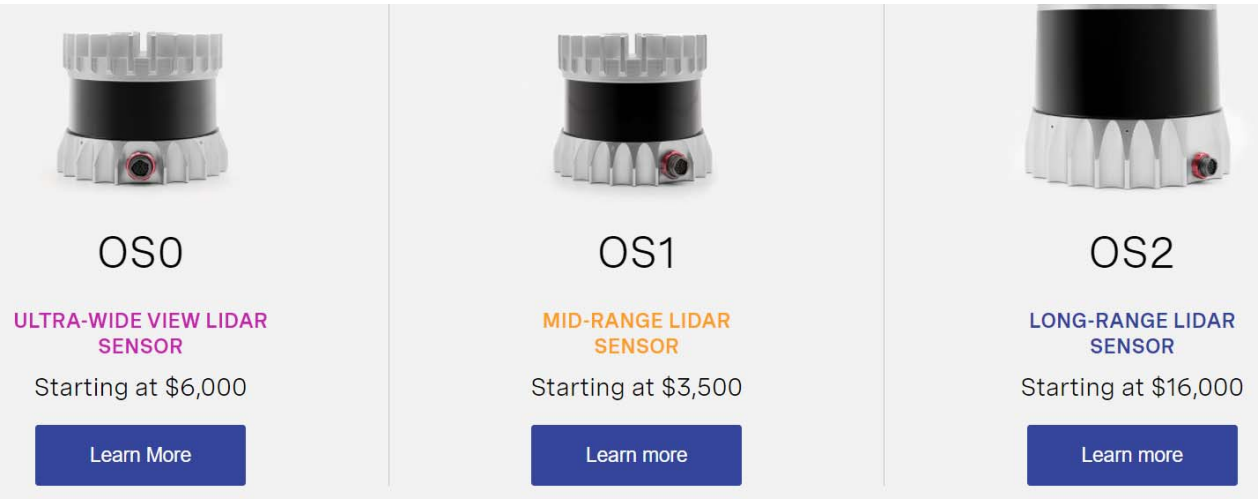
Small Size, Big Safety

Velodyne's smallest sensor brings new levels of versatility and affordability to 3D lidar perception. The compact, mid-range Velabit is highly configurable for specialized use cases and can be embedded almost anywhere within vehicles, robots, UAVs, and infrastructure. With the Velabit, the possibilities are endless.

Features:

Compact size of **2.4" x 2.4" x 1.38"**
Range up to 100 meters
60-º HFoV x 10-º VFoV
Highly configurable to support a range of applications
Class 1 eye-safe
903 nanometer
Available to customers mid-2020

Ouster https://ouster.com/



Ouster https://ouster.com/

- ES2 is the first high-resolution, long-range, solid-state digital lidar. Combining a VCSEL laser chip and a SPAD detector array raises the bar for resolution, range, and reliability. This is a sensor designed for those who want to reimagine what lidar can be simple
- Two chips + two lenses = Zero moving parts
- Size: 116x68x48 mm (89x68x48 mm:mounting 제외)



Designed for series production

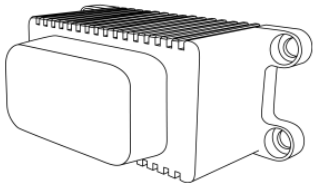
From our founding, we have designed our lidar technology with a solid-state architecture in mind. We're now realizing our original vision, with ES2 A-samples due in 2022 for SOP 2024.

Ouster https://ouster.com/



- ES2

ES2
True Solid-State Long-Range Lidar



SUMMARY
The ES2 is the first high-performance true solid-state digital lidar sensor. The solid-state electronic scanning ES2 features 200+ meter range, 0.1° horizontal and vertical angular resolution, and will have ISO 26262, ASIL-B, SIL-2 and ASPICE certifications. The sensor is designed for high-volume automotive, robotics, and industrial applications.

HIGHLIGHTS

- True solid-state digital lidar design
- Industry-leading resolution, reliability, and range
- Fixed resolution per frame
- Camera-grade ambient and intensity data
- Multi-sensor crosstalk immunity

OPTICAL PERFORMANCE

Range (80% Lambertian, >90% detection probability, 100 klx)	450 m
Range 10% Lambertian, >90% detection probability, 100 klx)	200 m
Minimum Range	0.25 m
Range Accuracy	±5 cm for lambertian targets, ±5 cm for retroreflectors
Precision (10% Lambertian target; 1 standard deviation)	0.25 m - 10 m: ± 0.5 cm 10 m - 150 m: ± 1.5 cm 150+ m: ± 3.0 cm
Range Resolution	0.3 cm
Vertical Angular Resolution	0.1°
Horizontal Angular Resolution	0.1°
Field of View	Vertical: 13° Horizontal: 26°
False Positive Rate	1/10,000
Frame Rate	10 - 30 Hz (configurable)
# of Returns	3 (first, strongest, last)

Datasheet

Ouster https://ouster.com/



- ES2

LASER

Laser Product Class	Class 1 eye-safe per IEC/EN 60825-1: 2014
Laser Wavelength	880 nm
Beam Divergence	0.03° (FWHM)

LIDAR OUTPUT

Connection	100BASE-T1 Automotive ethernet
Points Per Second	819,200
Data Per Point	Range, signal, reflectivity, ambient sunlight, channel, azimuth angle, timestamp
Timestamp Resolution	< 1 µs
Data Latency	< 10 ms

CONTROL INTERFACE

Connection	TCP and HTTP APIs
Time Synchronization	Input sources: • IEEE1588 Precision Time Protocol (PTP) • 802.11AS gPTP • NMEA $GPRMC UART message support • External PPS • Internal 10 ppm drift clock Output sources: • Configurable 1 - 60 Hz output pulse

MECHANICAL/ELECTRICAL

Power Consumption	12 - 18 W
Operating Voltage	9 - 26 V, 24 V nominal
Connector	Customizable to OEM specifications
Dimensions	Length x Width x Height: • With mounting screw holes: 116 mm x 68 mm x 48 mm • Without mounting screw holes: 89 mm x 68 mm x 48 mm

OPERATIONAL

Operating Temperature	-40 °C to +85 °C
Storage Temperature	-40 °C to +120 °C
Ingress Protection	IP68 (1m submersion for 1 hour, with I/O cable attached) IP69K (with I/O cable attached)
Shock	IEC 60068-2-27 (Amplitude: 100 g, Shape: 11 ms half-sine, 3 shocks x 6 directions) IEC 60068-2-29 (Amplitude: 40 g, Shape: 6 ms half-sine, 1,500 shocks x 6 directions)
Vibration	IEC 60068-2-64 (Amplitude: 3 G-rms, Shape: 10 - 1000 Hz, Mounting: sprung masses, 3 axes w/ 8 hr duration each)

Ouster https://ouster.com/



- ES2

Compliance	**Laser Safety:** • IEC/EN 60825-1:2014 Class 1 eye safe • US 21CFR1040: Complies with FDA performance standards for laser products except for deviations pursuant to Laser Notice No. 56, dated May 8th, 2019 **Product Safety: IEC/EN 62368-1:2014** **Functional Safety: ASIL-B** **EMC:** • FCC 47Cfr Part 15B, Class A • EN 55032: 2012/AC: 2013, Class A • IEC/EN 61000-4-3,4,5,6 • IEC/EN 61000-6-2:2005 Class A • IEC/EN 61000-6-4: 2007

WWW.OUSTER.COM

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Blickfeld https://www.blickfeld.com/ (Germany)

- Koito 협력사













Blickfeld https://www.blickfeld.com/ (Germany)

- CUBE 1 : MEMS LiDAR
- Cube 1 is a fully integrated scanning LiDAR sensor in a small form factor.
- Blickfeld‘s patented MEMS-based LiDAR technology enables a field of view of 70° x 30° and a maximum detection range of 250 meters. The sensor’s field of view, number of scan lines, and framerate can be flexibly configured via software, enabling a variety of application scenarios
- Size: 60x 80 x50 mm





Blickfeld https://www.blickfeld.com/ (Germany)

- CUBE 1 : MEMS LiDAR

OPTICAL PERFORMANCE*	
Detection range	5 m – 250 m (> 30m for 10 % reflectivity target, pixel-filling, 100 klux, 90 % detection rate, false positive rate < 0.2 %, 0.6° horizontal angular spacing)
Range resolution	< 1 cm
Range precision	< 2 cm (bias-free RMS, 10 m, 50 % reflectivity target)
Scan lines	> 500 scan lines per second (scan lines are software configurable, preprogrammed patterns include e.g. 70° x 20°, 0.4° horizontal point spacing, 50 scan lines, min. 10 Hz frame rate)
Maximum field of view (H x V)	70° x 30° (dependent on selected scan pattern)
Frame rate	1 – 30 Hz (dependent on number of configured scan lines)

*Measured at 25 °C, 60 % humidity, single return

LASER		
Laser class	Class 1, eye-safe (IEC 60825-1:2014, Ed. 3)	
Wavelength	905 nm	
Beam divergence	0.4°	Examples: 10 m: 0.07 m x 0.07 m 50 m: 0.35 m x 0.35 m 100 m: 0.7 m x 0.7 m

OUTPUT	
Connection	TCP/IP over Gigabit Ethernet
Data output per point	Azimuth and elevation angle, timestamp in µs per point, range, and Cartesian coordinates per return

Blickfeld https://www.blickfeld.com/ (Germany)

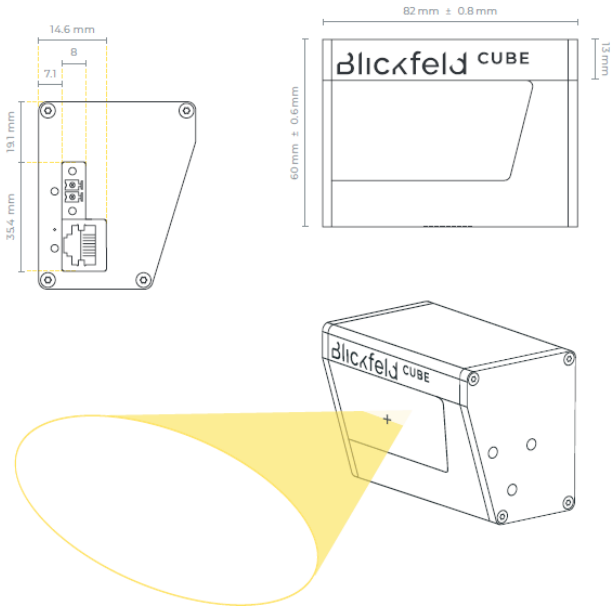
- CUBE 1 : MEMS LiDAR

CONTROL INTERFACE	
Configuration interface	Cross-platform graphical web interface
Control & stream interface	TCP connection with Blickfeld protocol. C++ library and Python package as client software interface
Time synchronization	NTPv4

MECHANICAL / ELECTRICAL	
Power consumption	Typ. 8.5 W (max. 15 W)
Operating voltage	Recommended: 12 V Absolute maximum ratings: Min.: 10 V Max.: 26 V
Dimensions (H x W x D)	60 x 82 x 50 mm
Weight	ca. 275 g
Data connector	Ethernet, RJ45
Power connector	Phoenix Contact 1817615 Compatible power supply connector: Phoenix Contact 1845219

OPERATIONAL	
Operating air temperature (angle mount attached)	-30 °C – 60 °C
Storage temperature	-30 °C – 60 °C
Humidity	85 % at 30 °C, non-condensing

DIMENSIONS



Innoviz Innoviz (Isreal)

- Israeli start-up founded in 2016
- **3D lidar based on MEMS mirror beam steering**
- Investment ($252 million USD) from Magna International and others
- Contract with BMW to supply the InnovizOneTM product in 2021 for BMW (**Magna** for **BMW**)
- InnovizOneTM is positioned as an automotive grade MEMS-based lidar module





Cepton https://www.cepton.com/

- FoV(HxW): 60x24
- Angular resolution(HxW): 0.25x0.25 deg
- Point image (HxW): 240x96=23,040(추정)





datasheet

Detection Range	200m
Weight	900g
FOV (HxV)	60 x 24 deg
Angular Resolution (HxV)	0.25 x 0.25 deg
Frame Rate	10 Hz
Size (WxHxD) mm	102 x 58 x 118mm
Power Consumption	10W
Range Resolution	2 cm (1σ)
Operating Voltage	10V-15V
Laser Wavelength	905nm
Laser Safety	Class-1 eye safe
Multi Return	Up to 2 returns
Vibrations/Shock	5Grms/50G 1000 shocks
Operating Temperature	-40°C ~ +85°C
Environmental Protection	IP67
GPS / Network	100 Mbps Ethernet, UDP Packet Synchronized Timestamps Position, Distance, Intensity
Platform Support:	Linux (Ubuntu 16.04), Windows 7+

Compatible with Vista-Edge™

2/05/2020 Businesswire

Cepton Expands With $50 Million Investment From Koito

Cepton announced today that it has raised over $50 million in Series C funding. This brings Cepton's total funding to nearly $100 million, providing a strong foundation to substantially ramp up its R&D efforts, expand its footprint in the automotive market and support major customers worldwide.

Pioneer Pioneer (Japan)

MEMS mirror system : Conventionally, a motor-driven 3D-LiDAR is needed to radiate laser light over a wide area. This means that it is difficult to make the system compact and lightweight due to the requirement of tons of components and the consequential high cost. This is why we have adopted a moter-less MEMS mirror system, which is capable of giving birth to a much more compact and lightweight 3D-LiDAR system.





Pioneer

Yun-Jeong Yang
Business Development Department

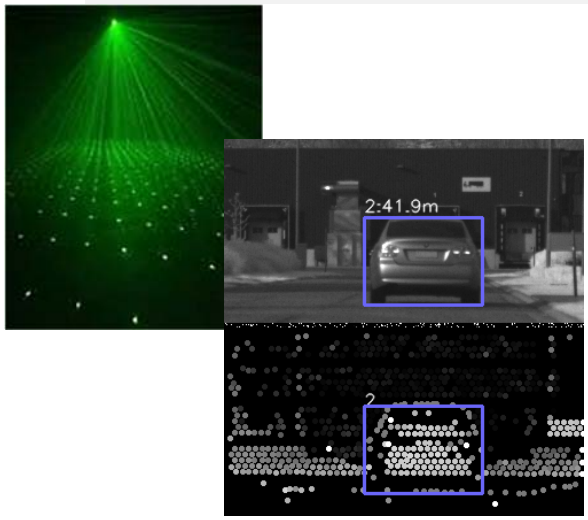
SMART SENSING INNOVATIONS

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XenomatiX Xenomatix (Belgium)

- VCSEL + CMOS(HD급 추정)
- Angular res.: 0.11x0.21 degree (X30), 0.21x0.42 degree (X60)
- Nb of beam: 14,750 (xenolidar <5,000)
- Size: 75X41X77 mm





	X30	X45	X60
HFOV	30°	45°	60°
VFOV	15°	22,5°	30°
Resolution	0,15°	0,22°	0,3°
Range (10% reflectivity)	200 m	100 m	50 m
Range (High reflectivity)	270 m	135 m	85 m
Power consumption	12W	12W	12W
LiDAR technology	XenomatiX Multi-beam	XenomatiX Multi-beam	XenomatiX Multi-beam
ASIL level	B	B	B
Laser classification	Class 1	Class 1	Class 1
Certification	AEC-Q100 grade 2	AEC-Q100 grade 2	AEC-Q100 grade 2

XenomatiX Xenomatix (Belgium)



A-sample of X-module (available end of 2020)

	X30	X60
HFOV	30°	60°
VFOV	10°	20°
Horizontal resolution	0,11°	0,21°
Vertical resolution	0,21°	0,42°
Range (20% reflectivity)	150 m	40 m
Range (High reflectivity)	200 m	50 m
Power consumption	20W	20W
Weight	360 g	360 g
Wavelength	940 nm	940 nm
Nb of beams	14,750	14,750
Dimensions	75 x 41 x 77 mm	75 x 41 x 77 mm
LiDAR technology	XenomatiX Multi-beam	XenomatiX Multi-beam
Laser classification	Class 1	Class 1
Certification	AEC-Q100 grade 2	AEC-Q100 grade 2

	X30	X45	X60
HFOV	30°	45°	60°
VFOV	15°	22,5°	30°
Resolution	0,15°	0,22°	0,3°
Range (10% reflectivity)	200 m	100 m	50 m
Range (High reflectivity)	270 m	135 m	85 m
Power consumption	12W	12W	12W
LiDAR technology	XenomatiX Multi-beam	XenomatiX Multi-beam	XenomatiX Multi-beam
ASIL level	B	B	B
Laser classification	Class 1	Class 1	Class 1
Certification	AEC-Q100 grade 2	AEC-Q100 grade 2	AEC-Q100 grade 2

XenomatiX Xenomatix (Belgium)

- 2차례 XenomatiX 방문 , CES2020 전시장 방문
- Xenomatix 사 magneti Marelli 공동개발 기사



Marelli and XenomatiX enter into a joint development agreement for LiDAR solutions

01/08/2020

Marelli and XenomatiX enter into a joint development agreement for LiDAR solutions

Las Vegas, 8 January 2020 - Marelli, a leading global tier-one automotive technology supplier, and XenomatiX, supplier of true "solid state LiDAR", announced today at CES® Las Vegas that they will enter into a technical and commercial development agreement in the autonomous driving field.

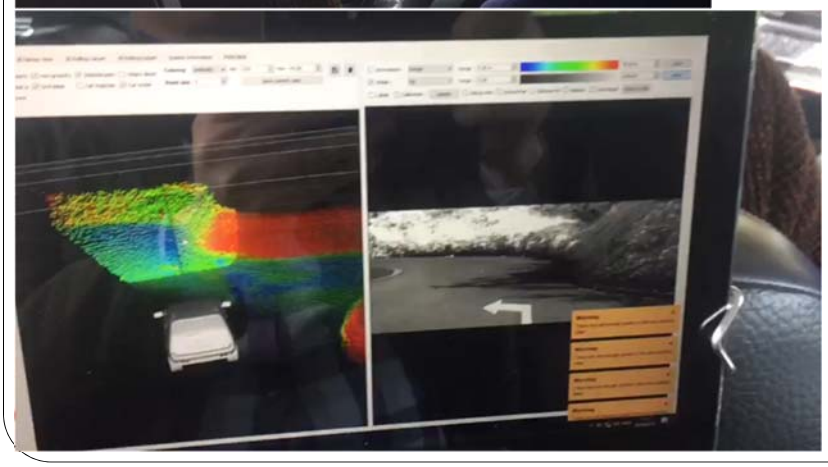
XenomatiX will provide Marelli`s

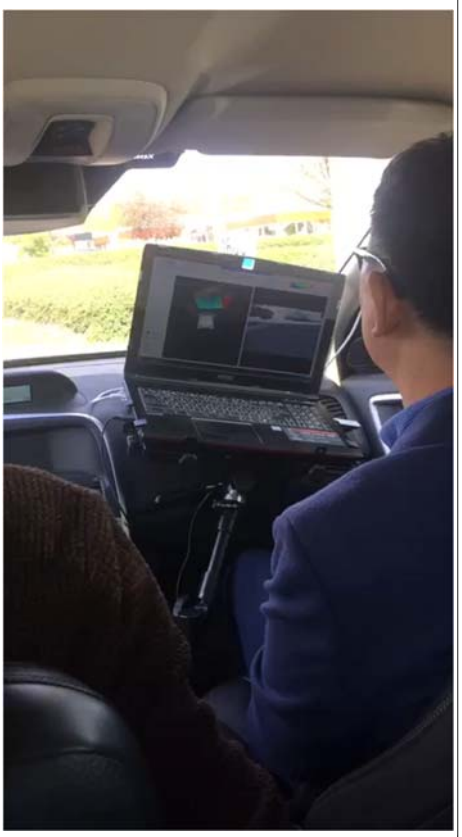




XenomatiX Xenomatix (Belgium)







LeddarTech LeddarTech (Canada)

- Canadian company founded in 2007
- Recently, launched the **LeddarTM Pixell 3D flash lidar** module, which is optimized for proximity detection and blind spot coverage in urban environments

Quanergy Quanergy (United States)

- Californian start-up founded in 2012
- developing the world's first optical phased array (OPA) 3D lidar for the automotive industry
- The company was valued at $1 billion USD in 2016
- Recent years, Quanergy has **shifted their attention to non-automotive markets**
- Question whether Quanergy has made the right technology choices in its OPA system or not: SiNx OPA platform might be **slow** and present **significant heat management issues**

Valeo Valeo (France)

- Valeo Scala® product by Valeo, a French automotive Tier 1 supplier, is already used in the Audi A8 production vehicle. This is a mechanically rotated lidar with few (x4) laser sources

Ibeo Ibeo (Germany)

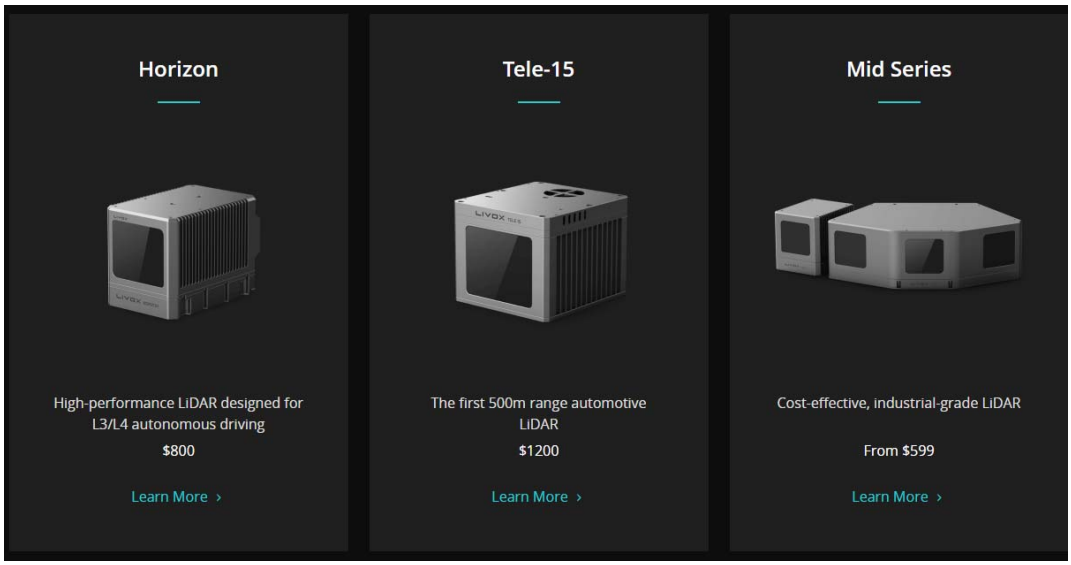
- Now developing a 905nm ToF lidar using a **VCSEL array** and an **SPAD array (12,800 pixels and emitters)**
  - boost the range to 250m even though the system architecture and components are similar to a flash lidar
  - A good cost-reduction path with **no scanning mechanism**
  - Working with their investors ZF Friedrichshafen AG.
  - CES2020 : 128ch. MEMS LiDAR 전시





Livox https://www.livoxtech.com/ (China)

- Set up by DJI, the famous Chinese drone maker
- ToF operating at 905nm, rotating prism
- Mid-40 lidar with 38.4deg FoV, 0.05-0.1 angular resolution and 260m range @80% reflective is priced at around 550-750 USD



SiLC https://www.silc.com/

- FMCW
- https://www.youtube.com/watch?v=viW5n



SiLC and Varroc Lighting Systems to Demonstrate Seamless Integration of LiDAR in Production Headlamp at CES 2020

SiLC also will demonstrate a 200+ meter LiDAR scan based on its second-generation FMCW silicon photonics 4D+ Vision Chip

Monrovia, Calif. and Plymouth, Mich., December 5, 2019 - SiLC Technologies, Inc., (SiLC), the leading provider of integrated single-chip FMCW LiDAR solutions, and Varroc Lighting Systems, a leading global supplier of innovative exterior vehicle lighting systems and electronics, today announced they will demonstrate a seamless LiDAR integration into a production automotive headlamp at CES 2020.

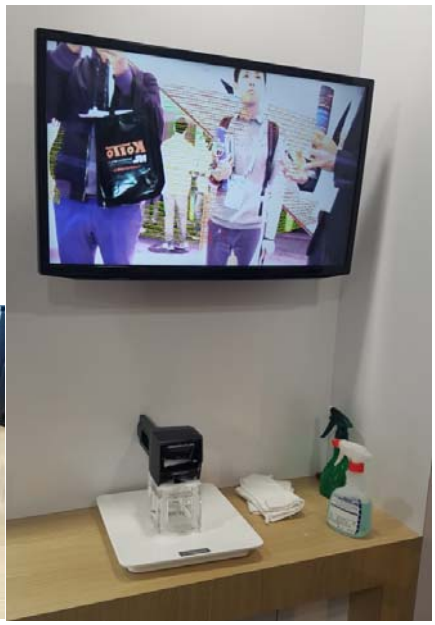


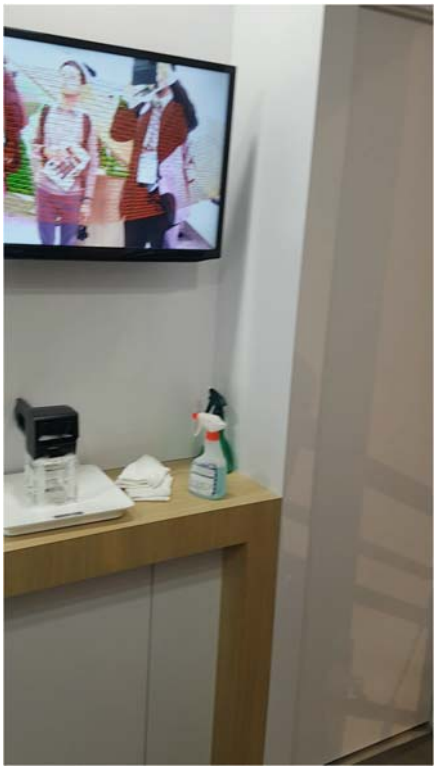
The Varroc Lighting Systems headlamp is based on a sophisticated production LED design and leverages four of SiLC's silicon photonics FMCW vision chips providing a full 20 x 80-degree field of view (FOV) per headlamp.

Koito (CES2020) Smart sensor: self-cleaning

- ADB
- Lidar Headlamp(cepton, Blickfeld)
- Dust 제거 데모







Path to Mass Production: Automotive Tier 1 Suppliers Enter the Game

- Major automotive OEMs prefer to obtain lidar technology for their autonomous production vehicles via trusted Tier 1 partners
- The general trend in the automotive industry is that Tier 2 suppliers develop new lidar technologies, while Tier 1 suppliers optimize these products for high-volume production.
  - Valeo Scala® technology was originally developed by Ibeo Automotive Systems GmbH. Valeo developed Ibeo's lidar into a mass-produced sensor for Audi.
  - Similarly, Innoviz has partnered with Magna, the Canadian Tier 1 supplier, and Magna will deliver automotive grade lidar modules to BMW.

Automotive Electronics Council(AEC) Q-100

The part operating temperature grades are defined below:

Grade 0:	-40°C to +150°C ambient operating temperature range
Grade 1:	-40°C to +125°C ambient operating temperature range
Grade 2:	-40°C to +105°C ambient operating temperature range
Grade 3:	-40°C to +85°C ambient operating temperature range
Grade 4:	0°C to +70°C ambient operating temperature range

Ref

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[2] H.W. Yoo, N. Druml, D. Brunner, C. Schwarzl, T. Thurner, M. Hennecke, G. Schitter OVE, “MEMS-based lidar for autonomous driving”, Elektrotechnik & Informationstechnik (2018) 135/6: 408–415.

[3] YUNG-PENG CHANG, CHUN-NIEN LIU, ZINGWAY PEI, SHU-MING LEE, YEONG-KANG LAI, PIN HAN, HSING-KUN SHIH, AND WOOD-HI CHENG, “New scheme of LiDAR-embedded smart laser headlight for autonomous vehicles”, Vol. 27, No. 20, September 2019, Optics Express

[4] Low Cost LiDAR System Design Considerations : https://www.youtube.com/watch?v=N7gnoxVrjR8

[5] SensL Solid State LiDAR design consideration : https://www.youtube.com/watch?v=npnAr1BlQhw

[6] The Next Generation of SPAD Arrays for Automotive LiDAR : https://www.youtube.com/watch?v=upBYO_KS_Is

[7] “LiDAR systems: costs, integration, and major manufacturers”, 05.03.2020 |Author /Editor:Seth Lambert/ Erika Granath, https://www.mes-insights.com/lidar-systems-costs-integration-and-major-manufacturers-a-908358/

[8] Dr Khasha Ghaffarzadeh, “Automotive Lidar Market: Battlefield of a Hundred Technology Suppliers”, Mar 16, 2020 https://www.idtechex.com/en/research-article/automotive-lidar-market-battlefield-of-a-hundred-technology-suppliers/20156

[9] Automotive Lidar 2020, http://www.automotivelidar.com/

Thank You

Next : Point Processing